

Ideas Parked — Not Yet Merged

Staging document; each entry marks its eventual destination

This document holds ideas discussed against Pattern Arithmetic v15.19, Inference, and Assembly, none of which have been written into any of those three files yet. Each entry states its status and its intended destination. Nothing here is settled; the point of this file is to let that be true without leaving Paper 1 mid-edit.

1 Destined for Pattern Arithmetic (untouched, remains open — now far more precisely characterized)

Mix across differing directions. Still open. No formula has changed, no route has been chosen. What changed is how precisely the gap itself can now be stated — an extended discussion worked through what Mix actually *is*, why several everyday-seeming counterexamples turn out not to be counterexamples, and split the open question into two genuinely separate parts that had been sitting together as one.

[**The reading, argued but not yet adopted as the paper’s own position.**] Mix is not usefully described as “multiple vectors becoming one vector.” That framing is general enough to make the gap look like an oversight. A better reading: Mix takes several reports already agreed to be answers to *one* shared question, and returns a verdict about that question. The shared $\hat{n}$ isn’t a boundary condition sitting on top of a more general operator — it is what makes “these are answering the same thing” true in the first place. In Mix’s already-defined domain, $\hat{n}$ itself was never plural; only χ ever varied.

[**Supporting evidence for that reading, already on file elsewhere in this project.**]

- Same-direction, opposite-phase, equal strength already returns the null state, black — not a fabricated compromise. Genuine contradiction is allowed to cancel to nothing, which only makes sense if agreement and disagreement were well-defined to begin with, which requires one shared axis.
- Unmixing is provably, structurally ill-posed (verified: two genuinely different input pairs land on an identical output). Metamerism was checked as independent real-world precedent for the same fact: many-to-one perceptual reduction is genuinely lossy, not a design flaw to route around.
- The paper’s own opening line already states the principle directly: “Two notes yield a chord, not a larger integer... the word ‘add’ names a different rule.” A chord is several notes sounding together *because* they already share one frame — a key, a scale. Not a generic combination of any two sounds.

[**Resolved, not open: the “we mix all the time” challenge.**] Four everyday examples were checked directly against this reading. Sugar dissolving in tea is one dial (sweetness) turning further along itself — never needed differing directions. Red plus green additively is already confirmed,

in the paper’s own words, to sit “at the same place” — same $\hat{n}$, different χ — already inside Mix’s settled domain; “a new reality” (yellow) is a new position within one shared phase-space, not evidence that different directions were combined. Layer-based paint software was checked directly and confirms the framework’s own distinction under different names: compositing separate layers is non-destructive and reversible (a bowl, $\oplus$, nothing discarded), while flattening is documented, across every source checked, as explicitly destructive and irreversible (Mix, $\boxplus$) — “once flattened, individual layers cannot be separated again.” “Unmerge them later” is only ever possible because the layers were never actually flattened; the moment they genuinely are, every source agrees: gone, matching unmixing’s own ill-posedness exactly. The one example that is *not* resolved — representing a whole painting (color, brushstroke, texture) as a single vector — turns out to be Gap 1 itself, restated as a question rather than evidence that it is already answered.

[New: the open question splits into two, not one.] Scope (different sorts entirely, like color versus sweetness) was already handled — route 1, refuse outright, a category error rather than a hard case. What remains splits further, for the case of one shared sort whose members sit at genuinely different positions on one sphere (rain and drought sharing a weather-concept sphere, the way red and green share a color one):

1. *Embedding validity.* Does geometric closeness on that shared sphere actually track real semantic agreement or disagreement at all? For hue, yes, by deliberate construction: the circle is built so that maximally different hues sit antipodally. For an arbitrary sphere of concepts, nobody has stated this. It is not given for free by placing two things on one sphere.
2. *The merge rule.* Only given a validated embedding does the original question — what should combining two positions produce — become well-posed at all. This is the question the existing paragraph already names; it was previously the whole of Gap 1, and is now understood to be its second half.

[Still confirmed not to help, checked directly, carried forward from prior sessions.] Magnitude behaving well (self-bounding near genuine opposition, verified via the law of cosines) does not license the direction that comes with it — a small, well-behaved magnitude can still carry an illegitimately-produced direction. The Hill form and Assembly’s Hill bijection both fix magnitude saturation only, verified twice, never direction. Bowls, per-channel scoring, space-time gating, and the candlestick merge rule are all genuine, useful ways of *not needing* a combined direction — none of them state what one would be.

[Open, unchanged in substance, changed in precision.] No merge rule exists for either sub-question above. The three routes on file — refuse outright, rotate onto a shared axis via a stated T first, or build a genuine two-stage operator — remain unchosen. The result of any eventual resolution is deterministic and reproducible, not random; what is missing is licensing, not predictability.

2 Destined for Pattern Arithmetic (advances an existing open item)

Location as a sort — a third candidate. The existing paragraph names two candidates and adopts neither. A third was tested here: dedicate separate, disjoint dimensions to content, to time, and to location, rather than letting any of them share an axis.

[Verified.] Two shifts acting through disjoint content dimensions commute exactly (checked numerically, not approximately: time-then-location and location-then-time landed on the identical

point to full floating-point precision). Two shifts forced to share a single content dimension do not commute, confirming the disjointness is what matters, not rotation itself.

[**Verified.**] Content stays recoverable after both shifts: projecting back onto the content dimensions alone reproduced the original direction closely, while the time-axis and location-axis each carried a clean, separately-readable amount of shift, with nothing blended between them.

[**Open, unchanged.**] The encoding itself — what specific rotation a given time value or location corresponds to — is not stated anywhere. The geometry now supports it cleanly; the mapping does not exist yet.

This is real, checkable content, not yet written as an addition to the existing paragraph.

3 Destined for Pattern Arithmetic (open item stays open, unchanged)

Reducing a path by merging close actors. The existing paragraph asks for three things: a stated distance, a merge rule, and a bound on how far a reduced path’s endpoint may drift from the unreduced one. Discussion here supplied a distance (see below) and partially addressed the merge rule, but only for magnitude, not for direction — and direction is the harder half this item is actually about.

[**Verified.**] Switching the strength calculation from the companion paper’s clip to Inference’s already-established Hill form, $r' = |z|/(K + |z|)$, changes nothing about the resulting direction $\chi' = \arg(z)$. Checked directly on two constructed sequences with identical content and opposite order: both gave the same angle under either formula. The Hill substitution fixes evidence-count blindness in strength; it does not touch the order-blindness this open item is actually asking about.

[**Open, unchanged.**] No merge rule for direction exists. No drift bound exists for either magnitude or direction. This item remains exactly as open as it was before this discussion, now with the magnitude half more precisely separated from the direction half.

4 Destined for Inference (new mechanism)

Candlestick summary: a richer merge rule for reference-atom building. Not a resolution of the Paper 1 item above — a different question. Paper 1’s open item is about *walking*: can a shortened path still be walked to roughly the same endpoint. This is about *reading*: how to summarize a group of atoms for comparison against a reference atom, which is Inference’s own established business (its reference-atom section), not *T*’s.

[**Verified.**] Open/close/high/low, computed the way a trading candle is, distinguishes exactly the case flat Mix could not. Two sequences, identical content, opposite order — naive Mix gave the identical angle (47.5) for both. The candle’s open and close swapped between them ($20 \rightarrow 160$ versus $160 \rightarrow 20$), correctly reading one as ending badly and the other as ending well. High and low stayed identical for both, correctly capturing that only the *range* was shared, not the *direction of travel*.

[**Settled, matches existing machinery.**] The scope rule — every atom belongs to exactly one time unit, no overlap between buckets — is not new; it is precisely the discrete-time-bucket metadata already defined on a walk’s own steps in the companion paper.

[**Open.**] This gives Inference a genuinely better reference-atom merge rule than flat Mix. It is not proposed as a fix for Paper 1’s walking-specific open item, which remains untouched above. Whether open/close/high/low itself needs a coherence check analogous to $\bar{R}_c$ has not been considered.

5 Destined for Inference (naming caution, not yet a mechanism)

“Inference Mandelbrot” is not Assembly’s Mandelbrot, and should not share its name. Assembly’s multi-zoom promotes an unordered thing (a settled camp) and is fully handled by the Hill bijection already verified there. Whatever would promote an *ordered* thing (a walked path, a movie, a time series) into one higher-level atom is a different, harder problem — it is the direction half of the path-reduction item above, restated from a different angle. **[Naming flag only.]** If this is ever built, it wants its own name; reusing “Mandelbrot” for it would be the same kind of collision already caught and corrected for “collapse,” “singularity,” and the quantum swap test.

6 Destined across three papers (new, split by piece)

Meta vectors: a second kind of vector alongside information vectors. Proposed: an information atom may carry two distinct kinds of vector. Information vectors are Axiom 7’s bundle v_i ’s, capped at $r_i \leq 1$, potentially several sorts τ across a whole record. Meta vectors describe the atom rather than being part of its content — three named so far: V_{derived} , link vectors (pointers between atoms, one- or two-directional), and a timestamp vector. Explicitly left open whether more meta-vector types are needed. Meta vectors are not capped at 1, and grouping several atoms’ meta vectors together on like terms is proposed as a distinct mechanism for building higher structures. This is one idea that lands in three different places once traced to where each piece actually belongs.

[Destined for Pattern Arithmetic.] The taxonomy itself — information vectors versus meta vectors as a formal, named distinction sitting on top of Axiom 7 — is structural, the same kind of addition Axiom 7 itself was. V_{derived} already lives there. A timestamp vector, if adopted, would be a third, genuinely different treatment of time from the two already on file above: distinct from the discrete-bucket metadata already defined on a walk’s own step, and distinct from the dedicated-axis encoding in the location-as-a-sort entry. Not yet clear whether more than one of the three should coexist, or which.

[Destined for Assembly.] Link vectors specifically. This is the mechanism the shelf’s own neighbour-graph note was already naming without a concrete form: “a data structure on seats after rest, not T .” Grouping link vectors on like terms is collecting edges into a graph, not merging them — gathering, in the $\oplus$ sense — so it carries none of the open questions below it.

[Destined for Inference.] The grouping mechanism itself: querying several atoms’ meta vectors, confirming they sit close together, and promoting one shared meta-vector up to the group as a whole. This is the space-time gating idea from earlier discussion, generalized into a standing principle. Timestamp and location get *used* here; they are only *defined* in Pattern Arithmetic above.

[Verified, carried forward as a caveat rather than a new finding.] Combining several atoms’ V_{derived} ’s stays safe only under the constraint the original already carries: auxiliary, never promoted to a trusted, walkable atom on its own. Checked directly that neither Inference’s Hill form nor Assembly’s Hill bijection touches this constraint — both fix magnitude only, never direction — so grouping V_{derived} ’s inherits that same limit rather than escaping it.

[Note added here.] Link vectors were separately discussed as one possible mechanism for a limited, honest form of “uncollapse”: not inverting Mix (still impossible, unaffected by anything here), but keeping an explicit, external pointer from a collapsed atom back to the bowl that produced it, created at collapse time, before the bowl itself might otherwise be discarded. This is navigation via a kept record, not reconstruction from the collapsed result — the distinction the white-light/prism discussion turned on, confirmed there to matter.

7 Considered and found to need nothing filed

For completeness: the reference-atom and precursor-witness mechanism, the bundle-vector / uncapped V_{derived} distinction from Mix, and the $\pi = (x; w_1, \dots, w_n)$ poem trajectory were all discussed at length and all turned out to already be fully present in Inference and Paper 1 respectively. Nothing from those threads is parked here, since there was nothing left unresolved to park.